SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Maha-GR-Align

AI-Assisted Drafting and Legal Alignment System for Maharashtra Government Resolutions

1. Introduction

1.1 Purpose

This document defines the software requirements for Maha-GR-Align, an AI-assisted system that helps Maharashtra Government officers draft, verify, and align Government Resolutions (GRs) with legal and procedural standards. It is written in simple language for students to understand and implement.

1.2 Scope

The system will:

- Help draft Government Resolutions using AI
- Automatically check references to old GRs, circulars, and court orders
- Detect conflicts with other departments' existing policies
- Ensure consistent Marathi–English legal terminology
- Force compliance with the Maharashtra Manual of Office Procedure template

Out of scope: Full legal advice, automatic approval of GRs, or replacement of human officers.

1.3 Dataset

Primary open dataset of historical Maharashtra Government Resolutions:

- https://gr.maharashtra.gov.in/1145/Government-Resolutions
- https://dte.maharashtra.gov.in/government-resolutions-orders-letters-circulars-e/
- https://github.com/orgpedia/mahGRs

Students should use this repository for training, testing, and evaluation of NLP models.

1.4 Definitions

Term	Meaning
GR	Government Resolution – official policy order issued by the state government
Manual of Office Procedure	Official handbook that defines the correct format of Maharashtra GRs

2. Overall Description

2.1 Product Perspective

Maha-GR-Align is a web-based AI tool that sits between desk officers and the official GR archives. It does not replace existing systems; it assists officers by providing drafts, checks, and warnings.

2.2 Main Users

- Desk Officers (main users who draft GRs)
- Legal Translators (Marathi–English)
- Administrative / Legal Reviewers
- IT / System Administrators

2.3 High-Level Features

1. Reference Tracking – extract and verify citations in the preamble

2. Conflict Detection – find overlapping or contradictory clauses from other departments
3. Bilingual Terminology Check – keep Marathi and English legal terms consistent
4. Template Enforcement – force correct structure (financial blocks, signatures, budget heads)
5. Web Dashboard – simple interface for drafting, reviewing, and exporting

3. Functional Requirements

3.1 Reference Parser Module

FR-1 The system shall extract all cited GRs, circulars, and court orders from a draft preamble.
FR-2 The system shall verify whether each cited document exists in the official archive / mahGRs dataset.
FR-3 The system shall highlight missing or incorrect references for the user.

3.2 Semantic Conflict Detector

FR-4 The system shall convert new draft clauses into vector embeddings.
FR-5 The system shall search the mahGRs vector database for semantically similar clauses from other departments.
FR-6 The system shall display potential conflicts with similarity score and source GR link.

3.3 Bilingual Terminology Module

FR-7 The system shall maintain a controlled bilingual (Marathi–English) legal glossary.
FR-8 The system shall flag any term that does not match the approved glossary.
FR-9 The system shall suggest the correct standard term when a mismatch is found.

3.4 Template Enforcement Engine

FR-10 The system shall enforce the structure defined in the Maharashtra Manual of Office Procedure.
FR-11 Mandatory sections: Preamble (Background → Reference → Resolution), Financial Sanction Block, Signature Headers, Budget Head (e.g., 2054-00-101-02).
FR-12 The system shall reject or warn when any mandatory section is missing or wrongly formatted.

3.5 User Interface & Workflow

FR-13 The system shall provide a web dashboard where officers can upload or type a draft.
FR-14 The system shall show side-by-side original text, AI suggestions, and validation results.
FR-15 The system shall keep an audit trail of all AI suggestions and human edits.
FR-16 The system shall allow export of the final draft as PDF / DOCX.

4. Non-Functional Requirements

NFR-1 Performance: Reference check and conflict detection for a typical GR shall complete within 30 seconds.
NFR-2 Security: Sensitive drafts shall not leave the on-premise / State Data Centre environment.
NFR-3 Usability: Desk officers with basic computer skills shall be able to use the system after a short training.
NFR-4 Reliability: System uptime target ≥ 95% during office hours.
NFR-5 Language Support: Full support for Marathi and English (Unicode).
NFR-6 Explainability: Every AI suggestion must show the source document or rule that triggered it.

5. Simple System Architecture

The system has five main layers:

1. Data Layer – mahGRs dataset + official archives (PDF/HTML → text → embeddings)
2. NLP Engine – Fine-tuned Indic models (MahaMarathi / LLaMA-style) + entity recognition
3. Vector Search – FAISS or similar for semantic similarity
4. Compliance Rules – JSON/YAML schema of Manual of Office Procedure
5. Frontend – Simple web dashboard for officers

6. Data Requirements

Primary Dataset: https://github.com/orgpedia/mahGRs

- Contains historical Maharashtra GRs in machine-readable form
- Students must download, clean, and convert to text / embeddings
- Additional data (if available): latest circulars, court orders, Manual of Office Procedure text

Students are expected to perform OCR, language segmentation (Marathi/English), and entity annotation as part of the project.

7. Constraints and Assumptions

- Internet access may be restricted; prefer on-premise models where possible.
- GPU resources for fine-tuning may be limited – use LoRA / smaller models.
- Human officers remain the final authority; AI only assists.
- Legal accuracy of AI output is not guaranteed – always keep human-in-the-loop.

8. Related Student Project Ideas (from the same proposal set)

Project 2 – AI Decision Intelligence Dashboards for HTE

Build dashboards + predictive models for enrolment, faculty, placement, and institutional performance using HTE institutional data.

9. Simple Acceptance Criteria (for Students)

1. Given a sample GR preamble, the system correctly extracts at least 80% of cited references.
2. Given two conflicting clauses, the system flags the conflict with a similarity score > threshold.
3. A generated draft contains all mandatory template sections.
4. Marathi–English term mismatches are correctly highlighted against a small glossary.
5. A working web prototype allows a user to upload text and see validation results.

10. References

- Dataset: https://github.com/orgpedia/mahGRs
- Maharashtra Manual of Office Procedure (official handbook)
- Original project proposal: Maha-GR-Align – AI-Assisted Drafting and Legal Alignment System
- Indic NLP models: MahaMarathi, IndicBERT, and similar open models

— End of Simple SRS —
Students may expand any section for their full project report.